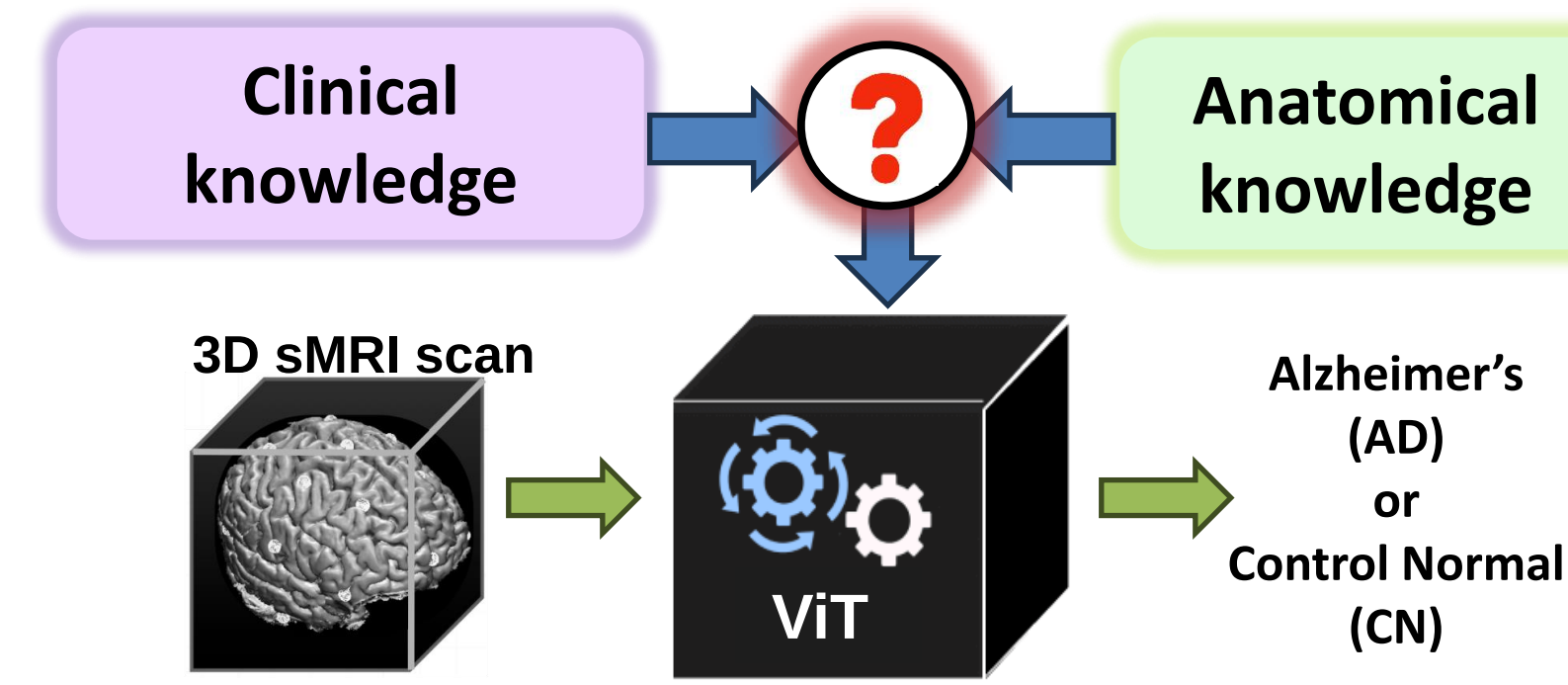
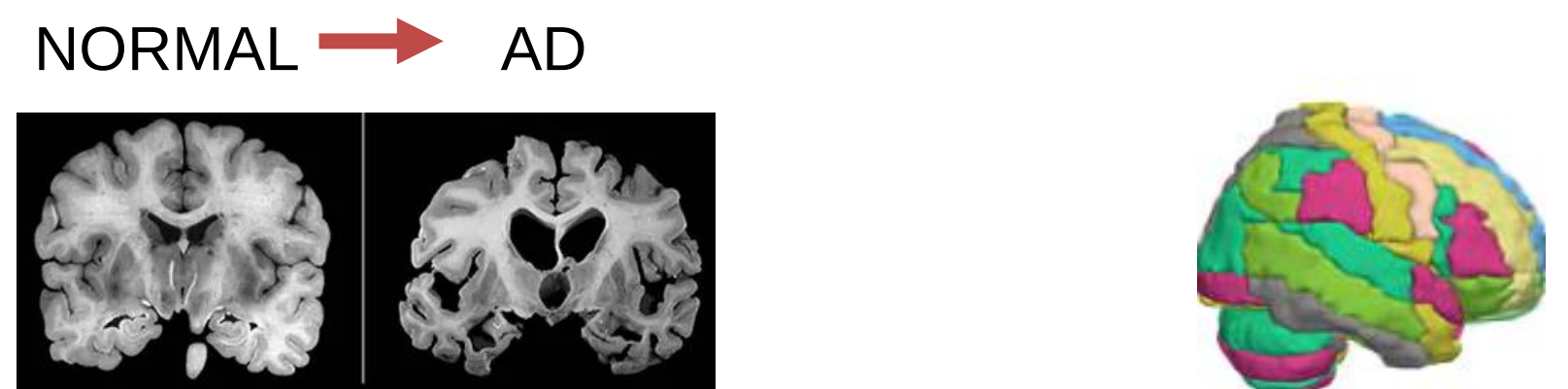


INTRODUCTION

- Dementia currently affects more than 60 million people worldwide, a figure expected to soar to 139 million by 2050.
- Clinically, Alzheimer's Disease progression is documented by atrophy in landmark regions (hippocampus, etc.) spread across brain.
- MRI processing technique localize various brain regions through atlases.

Can we incorporate crucial domain-specific knowledge, both clinical and anatomical, as an inductive bias for ViT-based models to learn effectively?



CONTRIBUTIONS

- Our contributions:
- **Relevance Augmented Self Attention (Rel-SA) module**, encoding clinical priors using Relevance Bias
 - Unifying two clinically validated brain atlases: 1) AAL3v1 and 2) JHU White Matter atlas
 - Qualitative Evaluation through **Leave-One-Out Analysis** and **Reverse Analysis**

MODEL ARCHITECTURE DESIGN

Rel-SA:

$$Rel-SA(Q, K, V) = \sigma \left(\frac{QK^T}{\sqrt{d}} + \phi(M; w_{high}, \alpha) \right) V$$

where, $Q = XW_Q$, $K = XW_K$, $V = XW_V$,
 $X \in \mathbb{R}^{N \times E}$, $\phi(\cdot) \in \mathbb{R}^{N \times N}$: *Rel - Bias*,
 $\sigma(\cdot)$: *softmax*

Score vector:

$$S(M; w_{high}, \alpha) = w_{high} \cdot M_{high} + w_{low} \cdot M_{low},$$

$$w_{low} = \alpha \cdot w_{high}$$

where, $S \in \mathbb{R}^{N \times 1}$, $M \in \mathbb{R}^{N \times 1}$, $\alpha \in (0, 1)$

Rel-Bias:

$$\phi(M; w_{high}, \alpha) = \beta \cdot \|S(M)S(M)^T\|_F$$

β is a hyperparameter

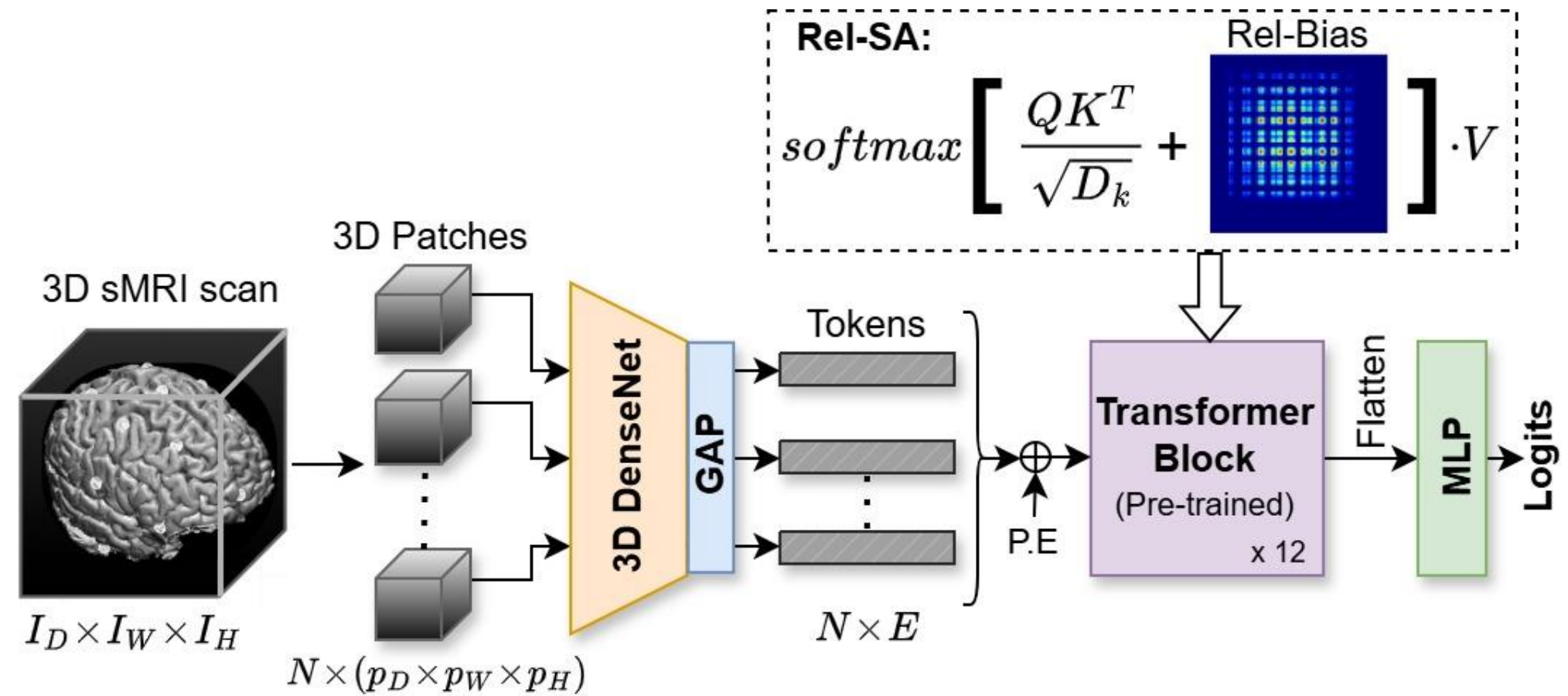


Fig 1: 3D Densenet+ViT Architecture with Rel-Bias in the Rel-SA module.

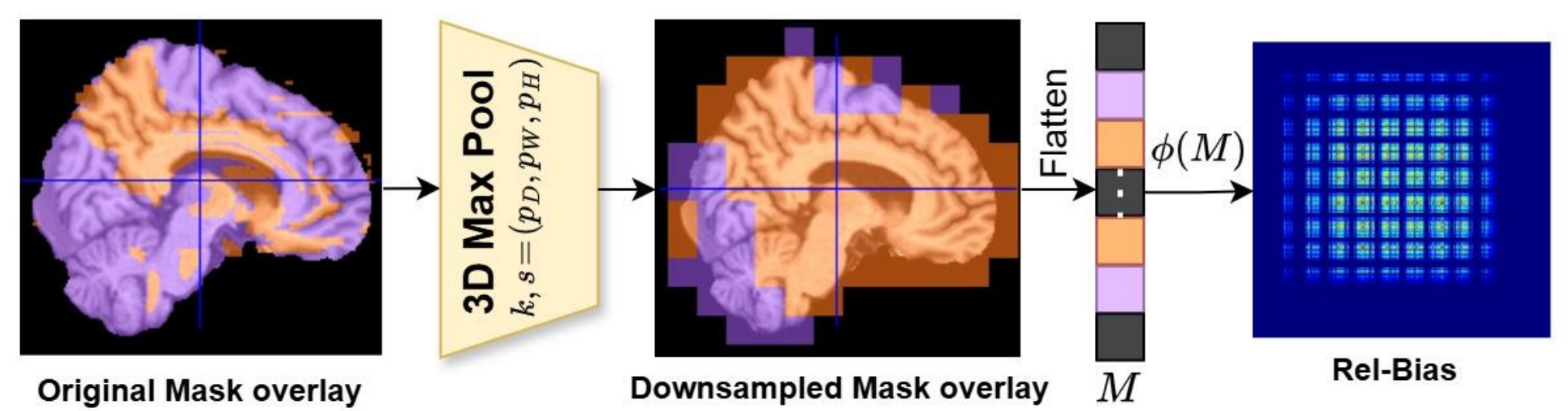


Fig 2: Rel-Bias calculation using relevance regions identified using Atlas.

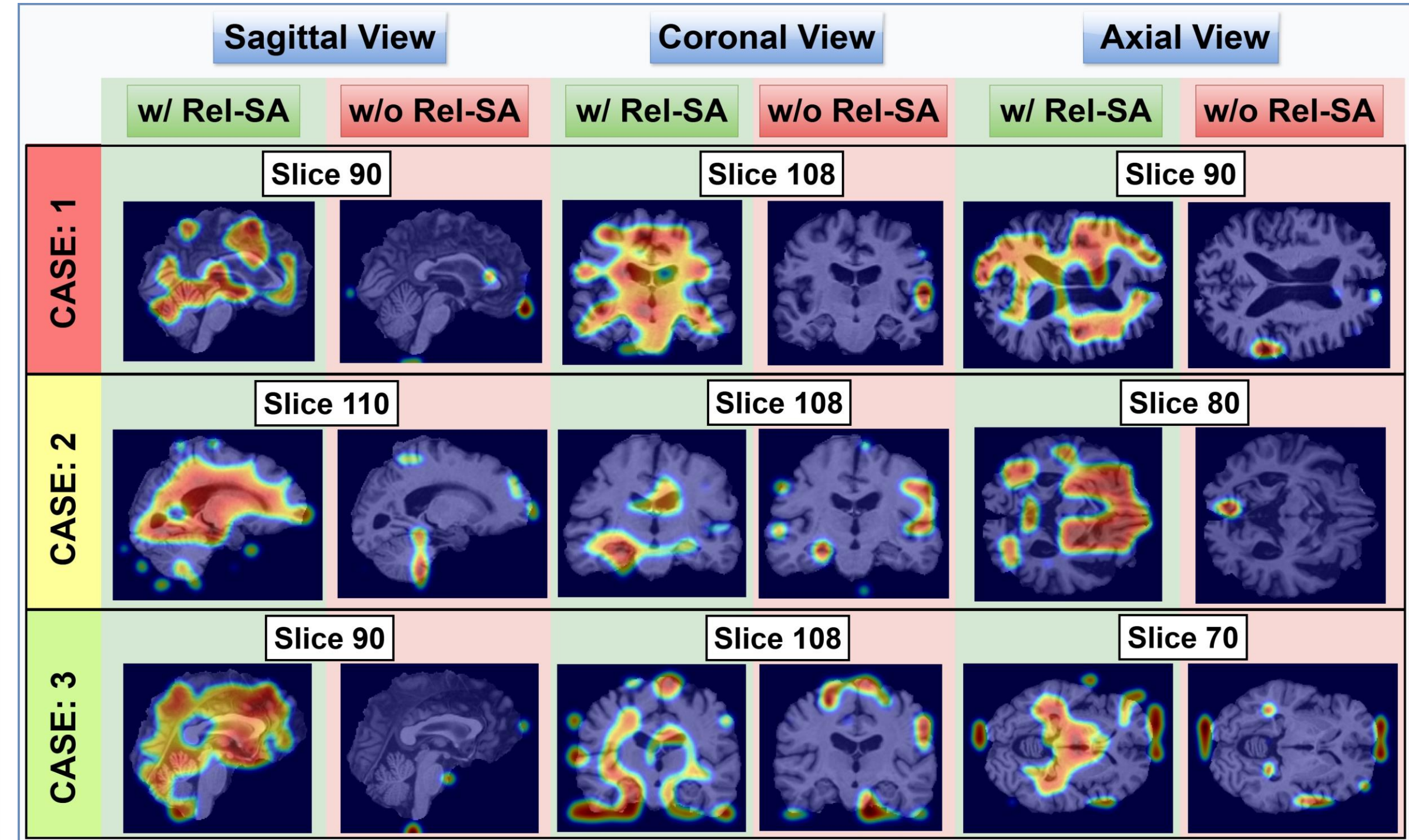


Fig 3: Case-wise Comparative Analysis of Attention Rollout visualizations. Case 1: GT = AD, both models predict AD; Case 2: GT = AD, Rel-SA predicts AD while vanilla ViT predicts CN; and Case 3: GT = CN, both models predict CN.

EXPERIMENTS

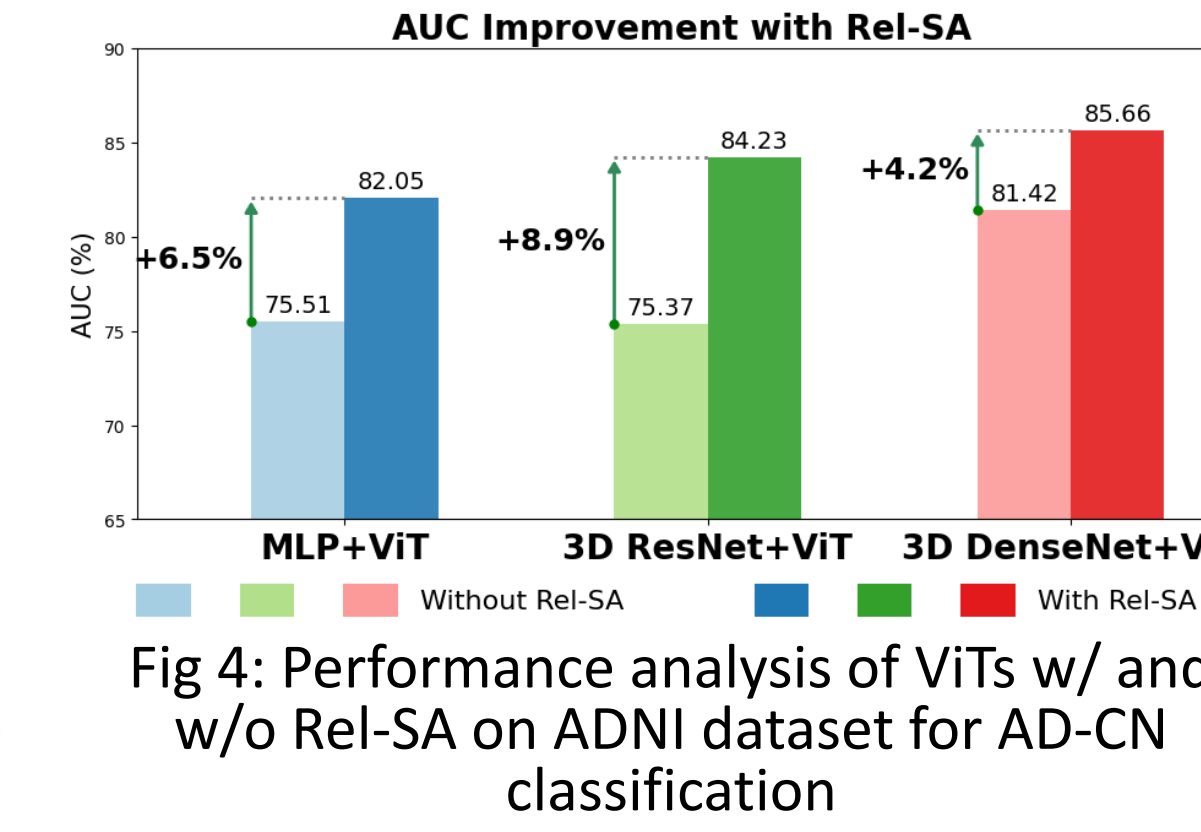
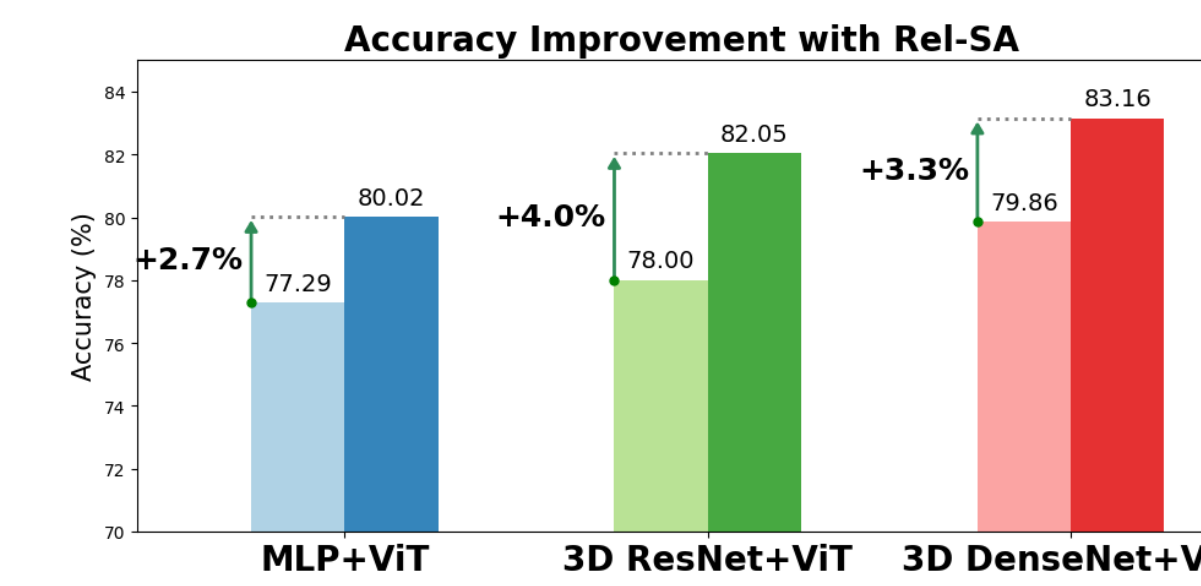


Fig 4: Performance analysis of ViTs w/ and w/o Rel-SA on ADNI dataset for AD-CN classification

Table 1: AD-CN classification analysis using Rel-SA on ADNI/AIBL

Dataset	Metric (%)	w/ Rel-SA	w/o Rel-SA
ADNI	Acc.	83.16 $\pm$ 0.54	79.86 $\pm$ 0.94
	AUC	85.66 $\pm$ 0.87	81.42 $\pm$ 1.83
AIBL	Acc.	87.40 $\pm$ 2.58	82.81 $\pm$ 1.55
	AUC	83.57 $\pm$ 2.25	81.42 $\pm$ 2.29

- Rel-SA's max boost to ViT-base: **Accuracy: ~4% & AUC: ~9%**
- Parametric overhead: just **24 additional parameters!**

CONCLUSION

- Our results depict that integrating clinical knowledge through Rel-SA can have a considerable impact both in terms of performance and faithful interpretations of the sMRI data consistent with Neuroscience literature.
- The key lies in leveraging domain knowledge as the right inductive bias.

POST-HOC ANALYSIS

Leave-One-Out Analysis

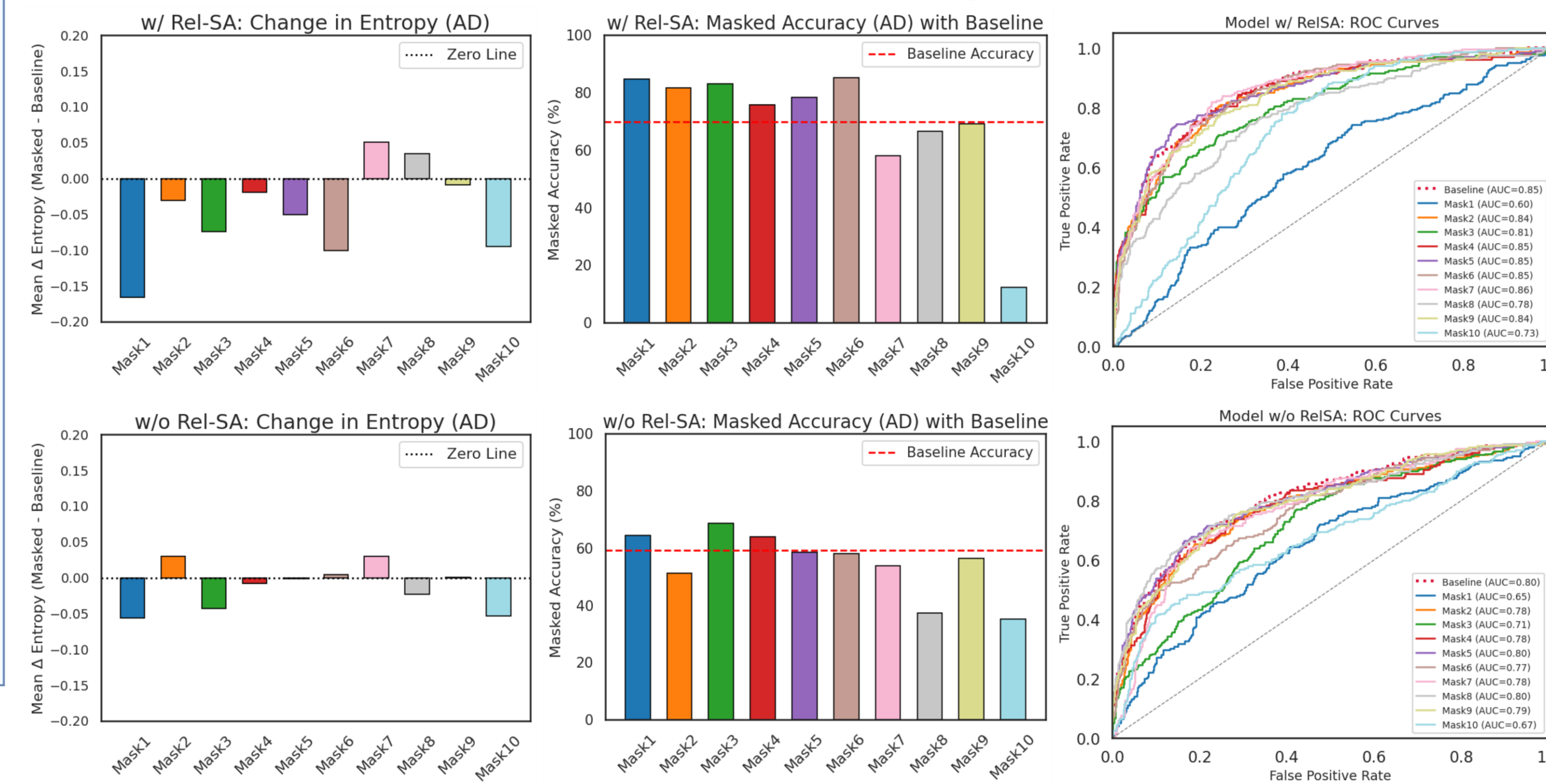


Fig 5: Leave-one-out Analysis of 3D Densenet+ViT model's performance w/ and w/o Rel-SA

Reverse Analysis

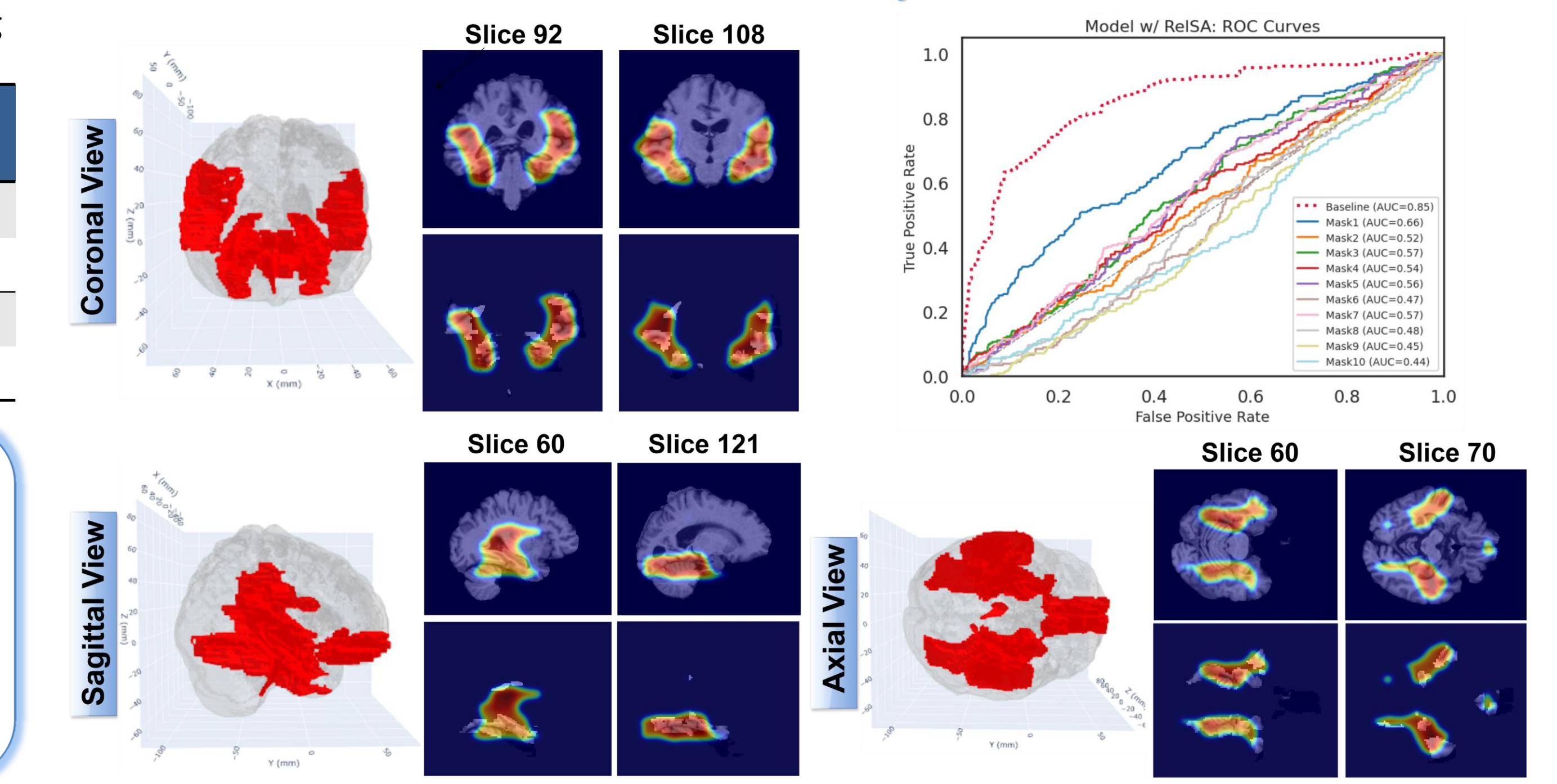


Fig 6: Reverse Analysis of 3D Densenet+ViT model's performance w/ and w/o Rel-SA using Mask1

CONTACT:

Madhumitha V
ai23resch11004@iith.ac.in
<https://madhuv29.github.io/>

Dr. Konda Reddy Mopuri
krmopuri@ai.iith.ac.in
<https://krmopuri.github.io/>

